

Fractal Information Theory Project

FIN ST263–ST277

Spontaneous Phase Orbits, Certified Selection,
and Trace-Class Obstructions

Analytical, exact, interval, and computational research report

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Repository `snapshot`. The audited tracked base is commit `10d0ccc19ac7b9101c7d382dd0859b352cec802f`. The working tree contains uncommitted and untracked research artifacts; reproducibility therefore depends on the accompanying SHA-256 manifest as well as the tracked base.

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Abstract

Programs ST263–ST277 attack three connected questions left by Release 10.74: whether a symmetry-broken state can arise without contradicting the strict D_{12} symmetry, whether the selector and refinement calculations can be certified more strongly, and whether the log-Haar scale plateau admits a mathematically controlled infinite completion.

The central constructive result is precise but conditional. The strict operator canonically determines a two-dimensional lowest positive Fourier eigenspace, hence a phase circle. The invariant algebra of D_{12} has no anisotropic scalar below degree twelve; its first such term is $\operatorname{Re} z^{12}$. Therefore a potential $-\kappa \operatorname{Re} z^{12}$ with $\kappa > 0$ has exactly twelve stable angular branches. A single realization breaks symmetry while their uniform ensemble restores it. However, neither κ , the nonzero radius, nor the realized phase follows from the strict operator. This is a minimal admissible mechanism for spontaneous selection, not a discharge of the selector obstruction.

Among the closure results, ST267 converts the previously numerical selector schedule into a unique global solution of the declared ten-knot discretized convex problem by an outward Krawczyk certificate for its active-set KKT system. ST268 proves the flat mean-zero affine connection natural under every exact linear-isometric refinement. ST269 certifies the full halfwidth-0.00070 nuisance cube and locates the fixed-resolution method boundary at the next declared radius. ST271 gives an exact continuum counterexample showing that two positive marginal Clifford gaps do not ensure joint uniqueness.

The strongest negative multiscale result is ST275: a nonzero exactly scale-invariant log-Haar distribution over all positive rates cannot have a trace-class heat completion, because its heat trace diverges logarithmically at zero. Cutoffs yield a controlled plateau with an explicit error bound but necessarily break exact scale invariance. No physical clock, length, Planck scale, spacetime, laboratory record, legacy-role transfer, Standard Model, gravity, or Theory-of-Everything closure is claimed.

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1 Scope and proof discipline

The strict working operator remains

$$A = sI - W, \quad W_{xx} = 0, \quad W_{xy} = K_{\text{strict}}(d(x, y)) \quad (x \neq y),$$

with $A\mathbf{1} = 0$ and $A \succeq 0$. The diagonal exclusion is part of the definition. The historical $K_{\text{legacy,ont}}$ remains an intermediate bridge kernel. No legacy physical role is transferred to the strict kernel in this report.

- **[PROVEN]** means an exact deduction or a stated finite outward certificate.
- **[CONDITIONAL]** means a theorem whose additional carrier, coefficient, cutoff, bath, or naturality premise is displayed explicitly.
- **[STRONG NUMERICAL EVIDENCE]** means a reproducible numerical result without a global certificate.
- **[BLOCKED]** means an evidentiary input is absent and is not replaced by simulation.

The nadsoliton is retained as the primordial fractal information in a solitonic state; no lower information layer is introduced. This ontological statement is not used as a premise in any mathematical proof below.

2 Executive result ledger

Pro-gram	Status	Result
ST263	[CONDITIONAL]	Strict lowest-mode circle; first anisotropic invariant $\text{Re } z^{12}$; twelve conditional stable phases.
ST264	[PROVEN]	Complete seven-generator polynomial equivariant module through degree 11.
ST265	[PROVEN]	Forty-four new interval boxes; no collision or stationary-rank loss on the targeted segment.
ST266	[PROVEN]	Rational global CPTP recovery optimum $3/8$ for a nonorthogonal measure-and-prepare channel.
ST267	[PROVEN]	Unique global optimum of the declared discrete selector-control problem.
ST268	[PROVEN]	Flat affine connection is unique and natural under exact linear-isometric refinements.
ST269	[PROVEN]	Complete 40^3 cover at 0.00070; fixed-resolution certificate fails at 0.00075.
ST270	[PROVEN]	Distribution-free simultaneous finite-confidence rectangles under clustered counts.
ST271	[PROVEN]	Circle of global Clifford minimizers despite two positive marginal gaps.
ST272	[PROVEN] + [STRONG NUMERICAL EVIDENCE]	Rigorous adaptive $s = 1/2$ enclosure; Chernoff- s optimizer remains numerical.
ST273	[CONDITIONAL]	Finite-controller reset cost $\beta W \geq h_2(p)$ when an uncertain record must be erased.
ST274	[CONDITIONAL]	Unlabelled-fiber naturality reduces refinement freedom from seven dimensions to one.
ST275	[PROVEN]	Finite-cutoff bound and impossibility of nonzero exact scale-invariant trace-class completion.
ST276	[PROVEN]	Frozen child-resolving instrument and record schema reconstruct the complement generator.
ST277	[BLOCKED]	No independently registered event record exists.

3 ST263–ST264: spontaneous phase orbits without a hidden selector

The smallest positive strict eigenvalue is

$$\lambda_1 = 0.7541211542070798$$

with real multiplicity two. Thus A canonically gives a carrier $E_1 \cong \mathbb{C}$, but it acts as $\lambda_1 I$ on that carrier and cannot choose a phase. Write the D_{12} action as

$$z \mapsto e^{2\pi i/12} z, \quad z \mapsto \bar{z}.$$

Theorem 3.1 (minimal anisotropic invariant). *Every real scalar D_{12} -invariant polynomial of degree below twelve is radial on E_1 . The first anisotropic invariant is $\operatorname{Re} z^{12}$. Conditional on a fixed radius $r > 0$ and $\kappa > 0$, the angular potential*

$$V_{\text{phase}}(z) = -\kappa \operatorname{Re} z^{12}$$

has exactly twelve stable angular minima.

The constructed probability branches are positive: their entries range from 0.0343435 to 0.1323231. Every branch has a reflection stabilizer of order two, whereas the equal ensemble of all twelve branches is uniform to 2×10^{-17} . This is the mathematically correct sense in which a symmetric law may permit individually broken realizations. It does not mean that symmetry itself is feedback: the radial instability and the anisotropic coefficient must still be generated by a dynamical law.

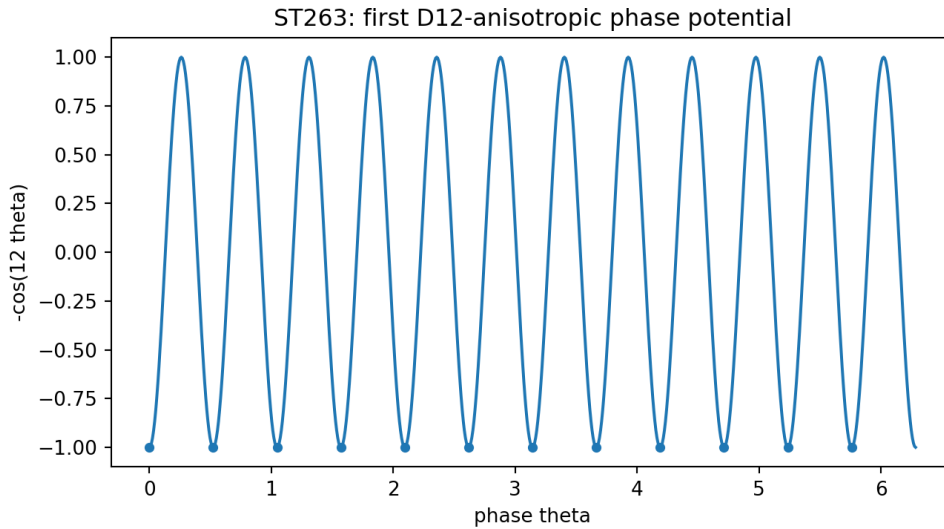


Figure 1: The first symmetry-allowed anisotropic phase potential has twelve equivalent minima.

ST264 sharpens the response classification. Through degree eleven, the real reflection-compatible D_{12} -equivariant module $F : E_1 \rightarrow E_1$ has the seven monomials

$$z, z|z|^2, z|z|^4, z|z|^6, z|z|^8, z|z|^{10}, \bar{z}^{11}.$$

The final generator is the first phase-sensitive one. Equivariance nevertheless implies $\operatorname{Stab}(z) \subseteq \operatorname{Stab}(F(z))$: processing can amplify a broken state, but cannot make its stabilizer smaller than that of its input. The missing scientific atom is now explicit: a strict source for the nonlinear coefficient, nonzero radius, and realized phase.

4 ST265: targeted event exclusion

The closest inherited nonfold seed pair, 20 and 23, was extended from step 8 to step 30. All 44 new roots have outward pseudo-arclength Krawczyk certificates. Their minimum new box clearance is 0.0183930. The smallest interval lower bound on the full row rank of the stationary Jacobian is

$$2.05253993945 \times 10^{-5} > 0.$$

The rank margin approaches a small positive value and then reopens; on the certified finite segment this is a near miss, not a collision or rank-loss event.

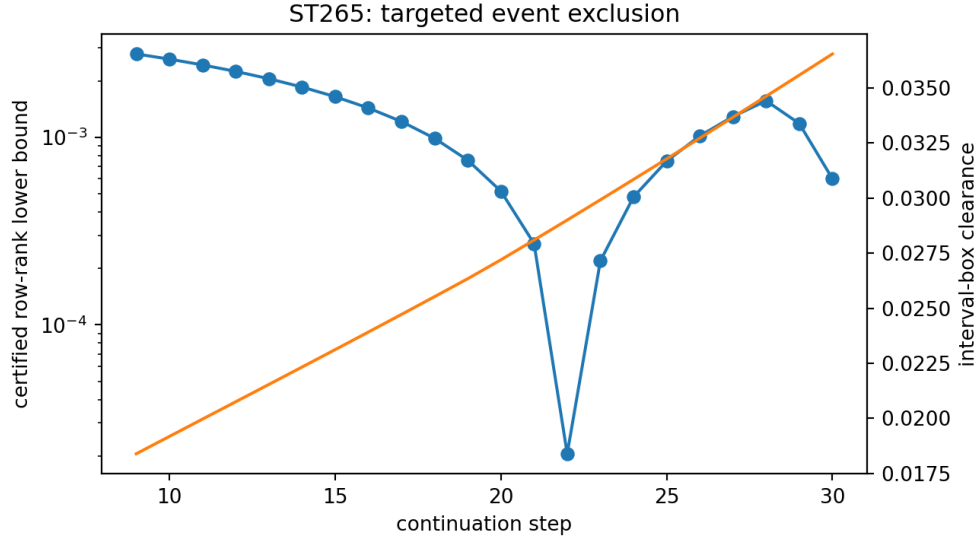


Figure 2: Certified stationary-rank lower bound and separation of the targeted branch pair.

No conclusion is made about later continuation, distant components, stability, or particles.

5 ST266: a recovery theorem outside the previous channel orbits

Consider the channel that measures the computational basis and prepares

$$\tau_0 = \begin{pmatrix} 1/2 & 3/10 \\ 3/10 & 1/2 \end{pmatrix}, \quad \tau_1 = \begin{pmatrix} 9/10 & 0 \\ 0 & 1/10 \end{pmatrix}.$$

The states are full rank and noncommuting. Optimizing recovery fidelity becomes binary state discrimination. Since $\Delta = \tau_0 - \tau_1$ has eigenvalues $\pm 1/2$, the rational Helstrom effect

$$M_0 = \begin{pmatrix} 1/10 & 3/10 \\ 3/10 & 9/10 \end{pmatrix}$$

attains discrimination sum $3/2$. The dual $\Gamma = (\tau_0 + \tau_1 + |\Delta|)/2$ dominates both states and also has trace $3/2$. Thus the global CPTP recovery optimum is

$$F_e^{\text{opt}} = \frac{1}{4} \frac{3}{2} = \frac{3}{8}.$$

The channel is nonunital, has Choi rank four, and has noncommuting outputs; these invariants exclude unitary equivalence to a Pauli mixture or qubit amplitude damping.

6 ST267: interval KKT closure of the selector model

The ten-knot schedule from ST252 has exactly one active inequality: the final negative slew face. Solving its 12-variable active KKT system gives

$$J_* = 0.4405461745248221, \quad \mu_* = 0.0577607478863 > 0.$$

An outward Krawczyk box with knot radius 2×10^{-9} and multiplier radius 2×10^{-8} has minimum inclusion margin 1.99997×10^{-9} . The nearest inactive slew slack is 0.0358324 and all box bounds are inactive. The objective Hessian has minimum eigenvalue 0.492701 at the root.

The entropy part is a sum of jointly convex Bernoulli Jeffreys divergences evaluated on affine variables; endpoint KL terms and the quadratic slew term make the complete objective strictly convex. Hence KKT sufficiency promotes the isolated root to the unique global minimizer of the exact declared ten-knot, 600-step discretized model.

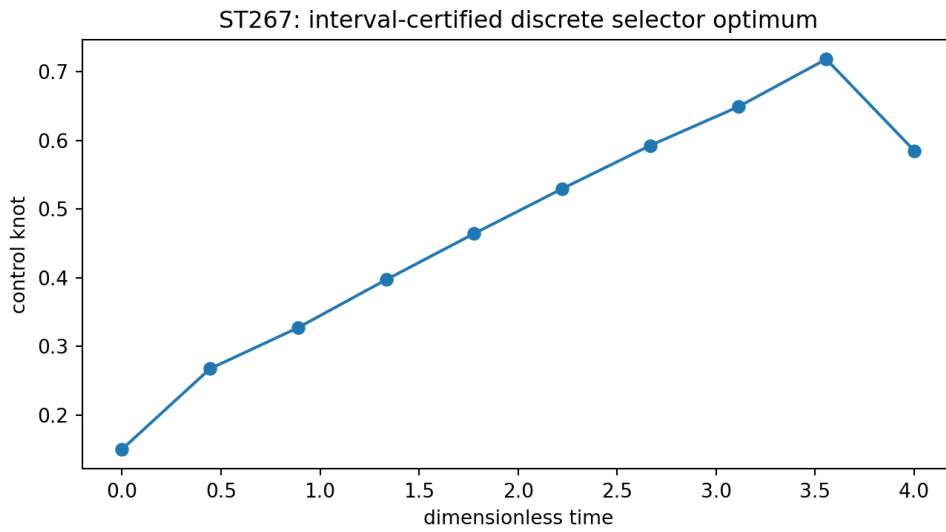


Figure 3: The unique interval-certified optimum of the declared discrete selector model.

This does not source the preferred endpoint, bath, rate constant, dimensionless clock, or physical energy unit.

7 ST268–ST270: refinement naturality, certificate boundary, and confidence

For an exact self-adjoint refinement

$$\tilde{A} = RAR^* \oplus B$$

with normalized linear isometry R , one has

$$g_{\tilde{A}}(Ru, Rv) = g_A(u, v), \quad R(\nabla_u v) = \tilde{\nabla}_{Ru}(Rv).$$

Thus the flat mean-zero affine Levi–Civita connection is unique and natural under the complete exact linear-isometric refinement class, including arbitrary positive complement B . This remains a carrier theorem, not spacetime geometry.

ST269 evaluates a predeclared fixed-resolution continuation campaign. All $40^3 = 64,000$ outward cells certify the nuisance cube of halfwidth 0.00070; the minimum margin is $2.12932459357 \times 10^{-4}$. At the next radius 0.00075, five of eight corner tiles fail. This refutes that proposed complete fixed-resolution cover but does not show root loss.

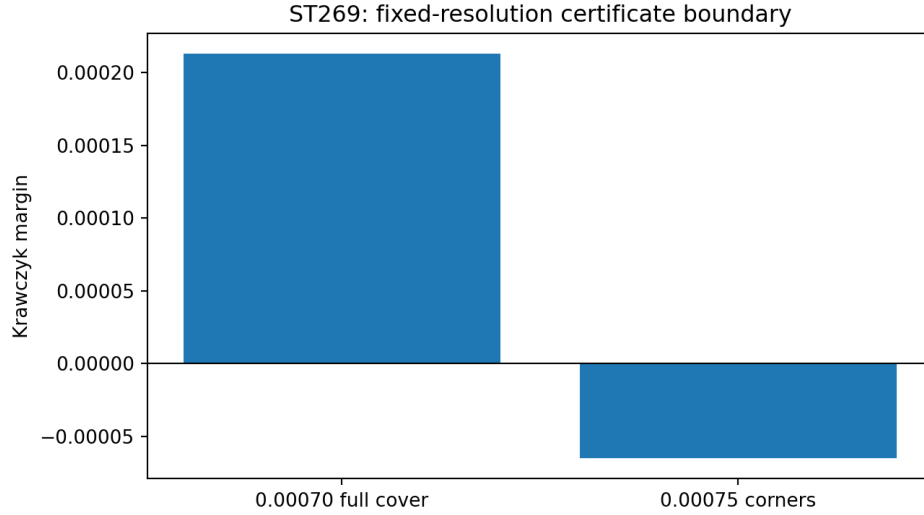


Figure 4: Positive full-cover margin at 0.00070 and the first failed fixed-resolution attempt.

ST270 treats every declared common-shock cluster as one independent bounded vector. A two-sided Hoeffding bound and a union bound across six modes produce a simultaneous 95% confidence rectangle at every admissible parameter. This is distribution-free but conservative; it relies on independent clusters, known attenuation, and the supplied count model.

8 ST271–ST272: falsifying a uniqueness heuristic and improving transfer bounds

Let

$$X_0 = \frac{3}{5}\sigma_z, \quad Z_0 = \frac{4}{5}\sigma_z.$$

Both targets have positive marginal sign gaps, but their Bloch columns are parallel. For every unit u_ϕ perpendicular to z ,

$$X_\phi = \frac{3}{5}\sigma_z + \frac{4}{5}u_\phi, \quad Z_\phi = \frac{4}{5}\sigma_z - \frac{3}{5}u_\phi$$

are anticommuting Hermitian involutions and attain the same global objective 2. Therefore the global minimizer set is a circle. Two marginal gaps alone are insufficient; the full-column-rank hypothesis cannot be removed.

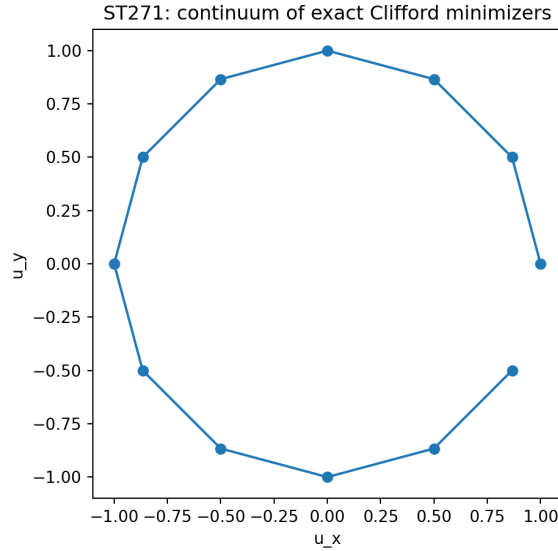


Figure 5: An exact circle of globally minimizing Clifford pairs.

For the belief-transfer operator, three adaptive rounds reduce the spectral-radius interval width from 0.0556162 to 0.0509913. The rigorous fixed- $s = 1/2$ rate interval is

$$[0.7129988031061385, 0.8228442593887783].$$

Intersecting it with the independent earlier certificate gives

$$[0.7129988031061385, 0.8198470892461787].$$

Finite-depth point collocation suggests $s \approx 0.5007394167$, but no all- s interval derivative certificate was produced; this candidate remains **[STRONG NUMERICAL EVIDENCE]**.

9 ST273–ST274: controller and refinement resources

A two-state controller implements an exact controlled-SWAP. If its program bit is known and uncomputed reversibly, no positive universal Landauer cost follows. If the controller retains an uncertain classical record $\text{diag}(1-p, p)$ and must be restored to a pure standard state in a bath, then

$$\beta W_{\text{reset}} \geq h_2(p).$$

At $p = 1/2$ the bound is $\ln 2$. The bath, record law, and Hamiltonian are supplied; FIN does not derive them.

ST274 adds one explicit categorical premise: each two-point refinement fiber is unlabelled, so independent local S_2 relabellings are natural automorphisms. They exchange aligned and crossed horizontal weights and force

$$p_d = q_d = \frac{W_d}{2}, \quad d = 1, \dots, 6.$$

The seven-dimensional ST259 cone collapses to the single ray $v \geq 0$ of vertical rates. Because v is invariant under all relabellings, naturality still does not select it. This is a one-modulus obstruction, not a unique fractal refinement.

10 ST275: controlled plateaus and a trace-class impossibility theorem

For density ρ and cutoffs $0 < \mu_{\min} < \mu_{\max} < \infty$, the log-Haar fiber contribution is exactly

$$d_{\text{fib}}(t) = 2\rho \left[\log(1 + e^{-2\mu_{\min}t}) - \log(1 + e^{-2\mu_{\max}t}) \right].$$

Its deficit from the ideal plateau obeys

$$0 \leq 2\rho \ln 2 - d_{\text{fib}}(t) \leq 2\rho\mu_{\min}t + 2\rho e^{-2\mu_{\max}t}.$$

Theorem 10.1 (trace-class obstruction). *A nonzero exactly scale-invariant log-Haar density on all positive rates does not define a trace-class heat completion for any $t > 0$.*

Indeed, the heat trace contains

$$\int_0^\infty e^{-2\mu t} \rho \frac{d\mu}{\mu},$$

which diverges logarithmically at $\mu = 0$. A trace-class completion must introduce an infrared cutoff or a density vanishing at zero. Either modification breaks exact scale invariance. Therefore a finite visible scale plateau is mathematically possible, but an exact infinite self-similar tower and trace-class dynamics cannot coexist in this model.

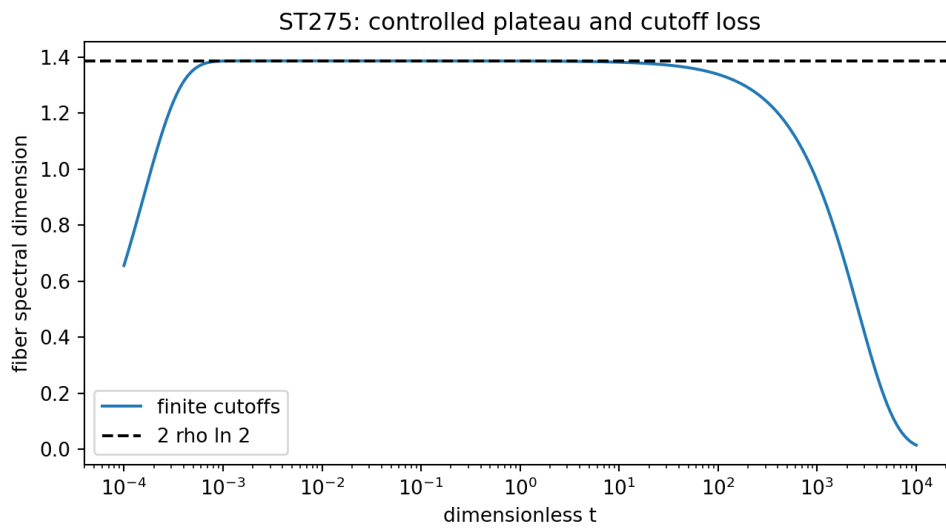


Figure 6: A controlled finite-cutoff plateau; both ends expose the necessary scale breaking.

Nothing in this theorem identifies the cutoffs with physical length, Planck scale, octaves, or legacy coupling formulas.

11 ST276–ST277: executable observation boundary

The frozen refinement instrument uses all 24 child-resolved preparations (x, a) and all 24 child-resolved effects (y, b) . The positive count probabilities determine every entry of $e^{-t\tilde{A}}$. The signed contrast

$$H_{\text{odd}}(y, x) = \frac{1}{2}(P_{y0|x0} - P_{y0|x1} - P_{y1|x0} + P_{y1|x1})$$

equals $S^*e^{-t\tilde{A}}S = e^{-tB}$. With calibrated t and the stated principal-log condition,

$$B = -t^{-1} \log H_{\text{odd}}.$$

The numerical replay error is below 10^{-14} .

The record schema freezes timestamps, preparation and effect labels, counts, run and calibration hashes, provider, registrar, analyst, and holdout status. Provider, registrar, and analyst must be distinct. ST277 remains **[BLOCKED]**: no independently registered raw event record exists, so no empirical claim is made.

12 Falsification ledger and surviving interpretation

Candidate claim	Falsification attempt	Surviving statement
A selects a phase	Functional calculus on the degenerate E_1 sector	A selects the carrier, not a point on its phase circle.
Symmetry breaking must be inserted as a direction	Invariant-theory search	The minimal anisotropy has forced degree 12, but its coefficient and realized branch are unsourced.
Two spectral gaps imply joint Clifford uniqueness	Parallel full-gap targets	False: an exact circle of global minimizers exists.
Locality selects refinement	Independent fiber relabelling	Freedom shrinks to one positive vertical rate, not zero.
Exact log-Haar flow gives an infinite physical fractal	Heat-trace audit near $\mu = 0$	False without a cutoff or vanishing density; either breaks exact scale invariance.
Executable specification is evidence	Custody and record audit	False: ST277 remains blocked.

The deepest statement surviving this batch is

A canonically determines a symmetric carrier and its allowed response algebra,
 A does not determine the nonlinear source, realized branch, or physical scale.

This refines the information-first intuition. Representation-independent structure may be encoded by the carrier, group action, operators, preparations, effects, and transformation laws rather than by the spectrum alone. Different observational layers can hide complement generators, but an explicitly richer instrument can recover them. Information is therefore not shown to propagate “without any carrier”; instead, the same operational structure may be instantiated in different carriers once a representation functor and preserved observables are supplied.

13 Recommended programs ST278–ST292

- ST278.** Derive the coefficient and radius of the $\operatorname{Re} z^{12}$ potential from one strict nonlinear interaction, or prove a source obstruction.
- ST279.** Compute a Hilbert basis for the complete nonlinear D_{12} -equivariant module beyond degree eleven.
- ST280.** Continue seed 23 only through a certified minimum-rank event bracket, with an explicit finite stop.
- ST281.** Generalize the exact measure-and-prepare recovery theorem to arbitrary rational binary qubit outputs.
- ST282.** Independently replay ST267 with rational or arbitrary-precision interval arithmetic.
- ST283.** Test affine-connection naturality under nonlinear refinement embeddings or construct a counterexample.
- ST284.** Adaptively subdivide failed 0.00075 nuisance cells to separate resolution loss from a root boundary.
- ST285.** Replace Hoeffding rectangles with exact-mixture or empirical-Bernstein clustered confidence sets.
- ST286.** Classify higher-dimensional joint Clifford nonuniqueness beyond the qubit rank-one counterexample.
- ST287.** Certify the global Chernoff- s optimizer with interval derivatives and complete s coverage.

- ST288.** Construct an autonomous finite controller and distinguish catalytic from resetting costs.
- ST289.** Impose coassociativity on $12 \rightarrow 24 \rightarrow 48$ unlabelled-fiber refinements and classify surviving vertical-rate moduli.
- ST290.** Classify near-Haar trace-class densities and prove the optimal plateau-width versus trace-class tradeoff.
- ST291.** Build an executable validator for the frozen ST276 instrument and custody schema.
- ST292.** Resume empirical validation only after independently registered raw events exist.

ST278 is the decisive theoretical continuation. If the degree-twelve coefficient and a nonzero radius can be derived from a strict interaction without already importing a broken state, FIN will have an internal symmetry-breaking dynamics, although it will still not choose one realized branch deterministically. ST290 is the decisive multiscale continuation: it should quantify how long an approximately scale-invariant visible plateau can coexist with trace-class completion.

14 Reproducibility and conclusion

The executable source `fin_st263_st277_research.py` writes the aggregate JSON, CSV ledger, fifteen per-program packets, and eight figures. The independent live replay `test_fin_st263_st277.py` checks packet hashes, the invariant module, all 44 continuation inclusions, rational recovery duality, the live KKT residual and convexity, the 64,000-box cover count, confidence construction, Clifford counterexample, adaptive transfer narrowing, refinement intertwining, trace bound, and operational reconstruction.

The batch produces a substantive but bounded advance. FIN now has an exact minimal mathematical template for twelve-branch spontaneous selection and a proof that symmetry can be restored at the ensemble level. It does not yet have the strict nonlinear law that makes this template inevitable. It also has a controlled interpretation of visible fractal scale plateaus, together with a proof that an exact infinite log-Haar tower is incompatible with trace-class heat dynamics. Consequently, neither spontaneous selection nor fractal layering may be promoted from a conditional mathematical mechanism to observed physics without the missing source and operational bridges.